Dimensional Modeling Concepts

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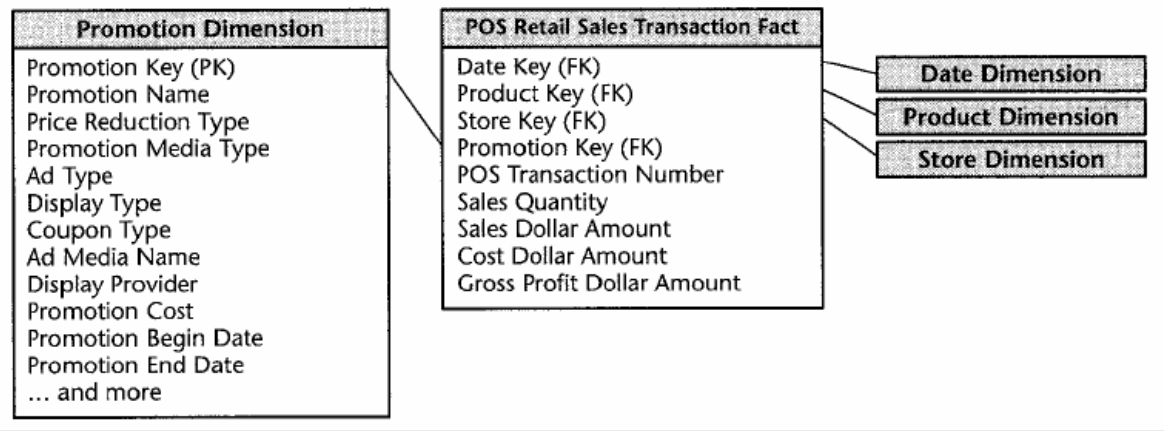


More Design Concepts

- Degenerate dimension
- Role-playing dimension
- Fact-less fact table
- Surrogate-Key
- Snowflake schema
- Multiple fact tables
- Different types of fact table
- Etc.

Degenerate Dimension

- Degenerate dimensions(that is, dimension keys without corresponding dimension tables) in fact tables
- Degenerate dimensions often play an integral role in the fact table's primary key.
- Operational control numbers such as
  - order numbers,
  - invoice numbers
- Those numbers usually give rise to empty dimensions and are represented as degenerate



Degenerate Dimension

Degenerate dimensions are dimensions without dimension tables

=> Here, Transaction Number

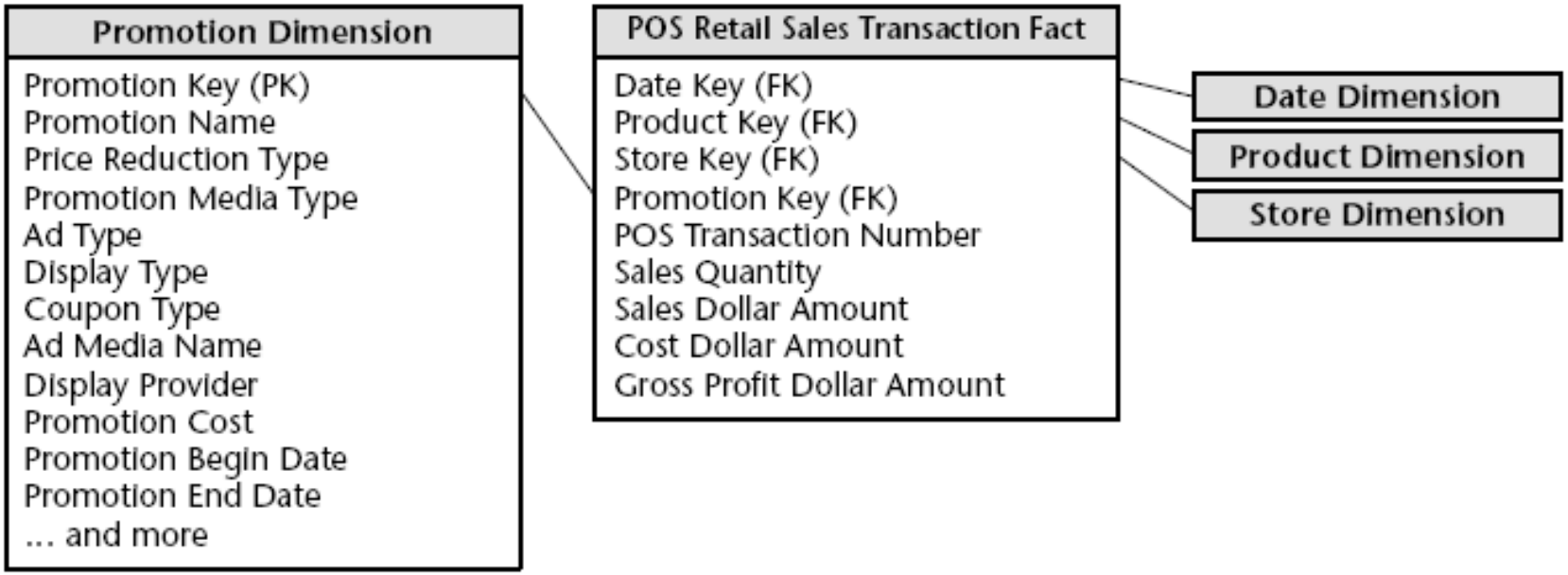
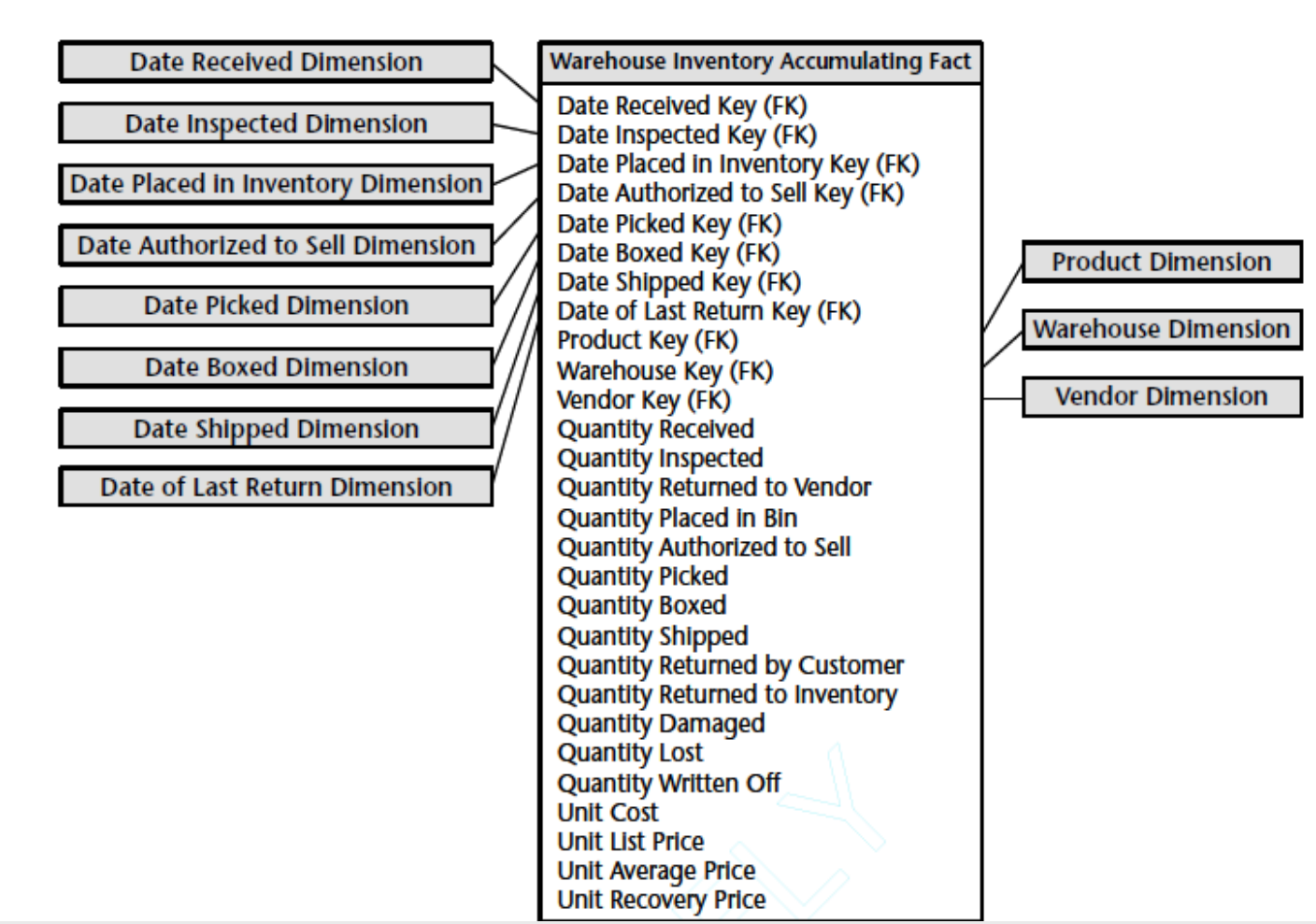


Figure 2.9 Promotion dimension in the retail sales schema.

Role-Playing Dimension

- **Role Playing dimension refers to a dimension can play different roles in a fact table depending on the context.**
- **For example, the Date dimension can be used for the ordered date, scheduled shipping date, shipment date, and invoice date in an order line fact.**
- **Each of date should be a foreign key in the fact table**
- **We cannot literally join to single date dimension table. Create the illusion of two independent date table by using views**
- **A *role-playing* dimension is a dimension in a star schema that has one physical representation (one underlying physical table) but is referred to more than once (different roles) by the fact table.**

Role-Playing Dimension



FACTLESS FACT TABLES

- **FACTLESS FACT TABLES**
  - **Event tables often have no obvious numeric facts (measures) associated with them**
  - **No facts (Measures) in the fact table**
  - **Used as a method for recording events in a data warehouse when there is no natural numeric measurement associated with the event**

Factless fact table

Event tables often have no obvious numeric facts (measures) associated with them

=> Factless Tables

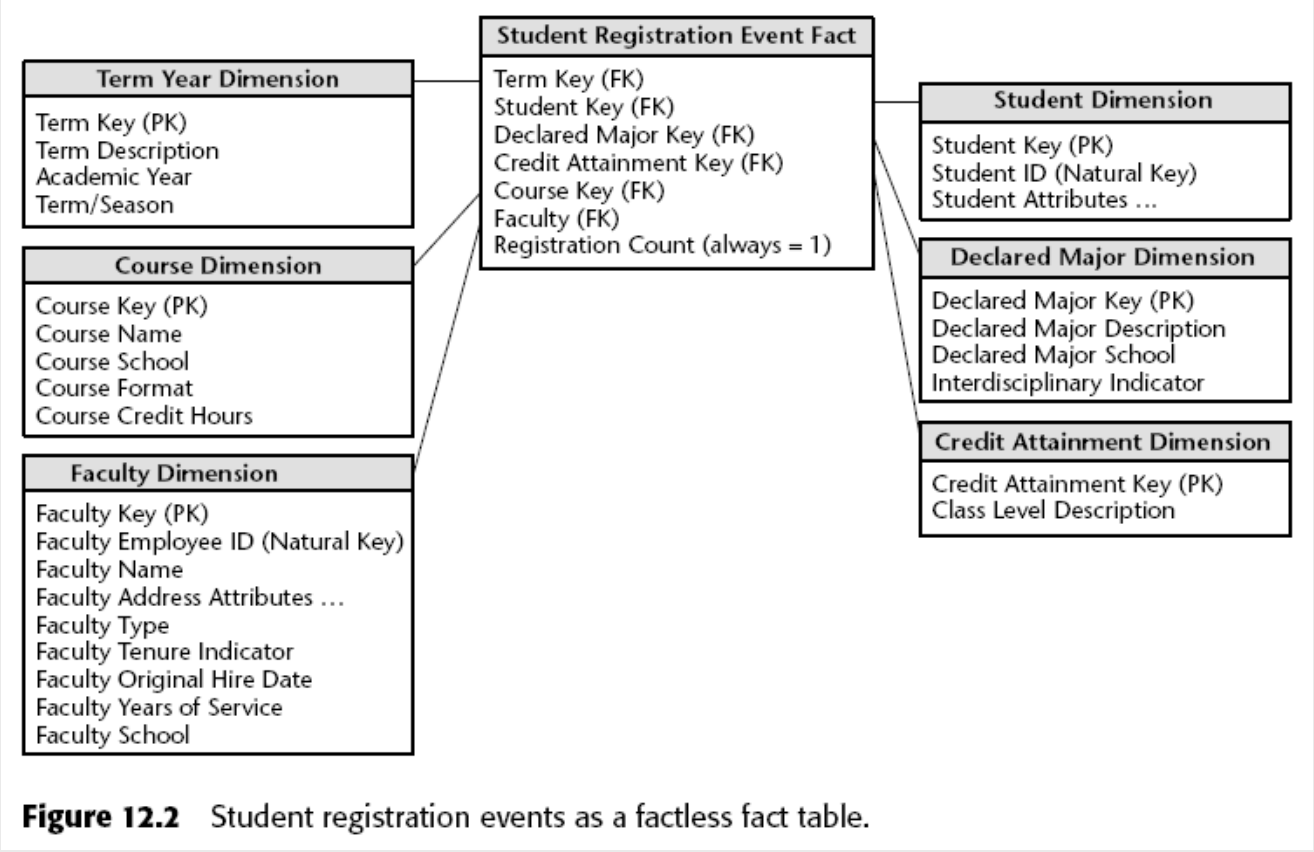
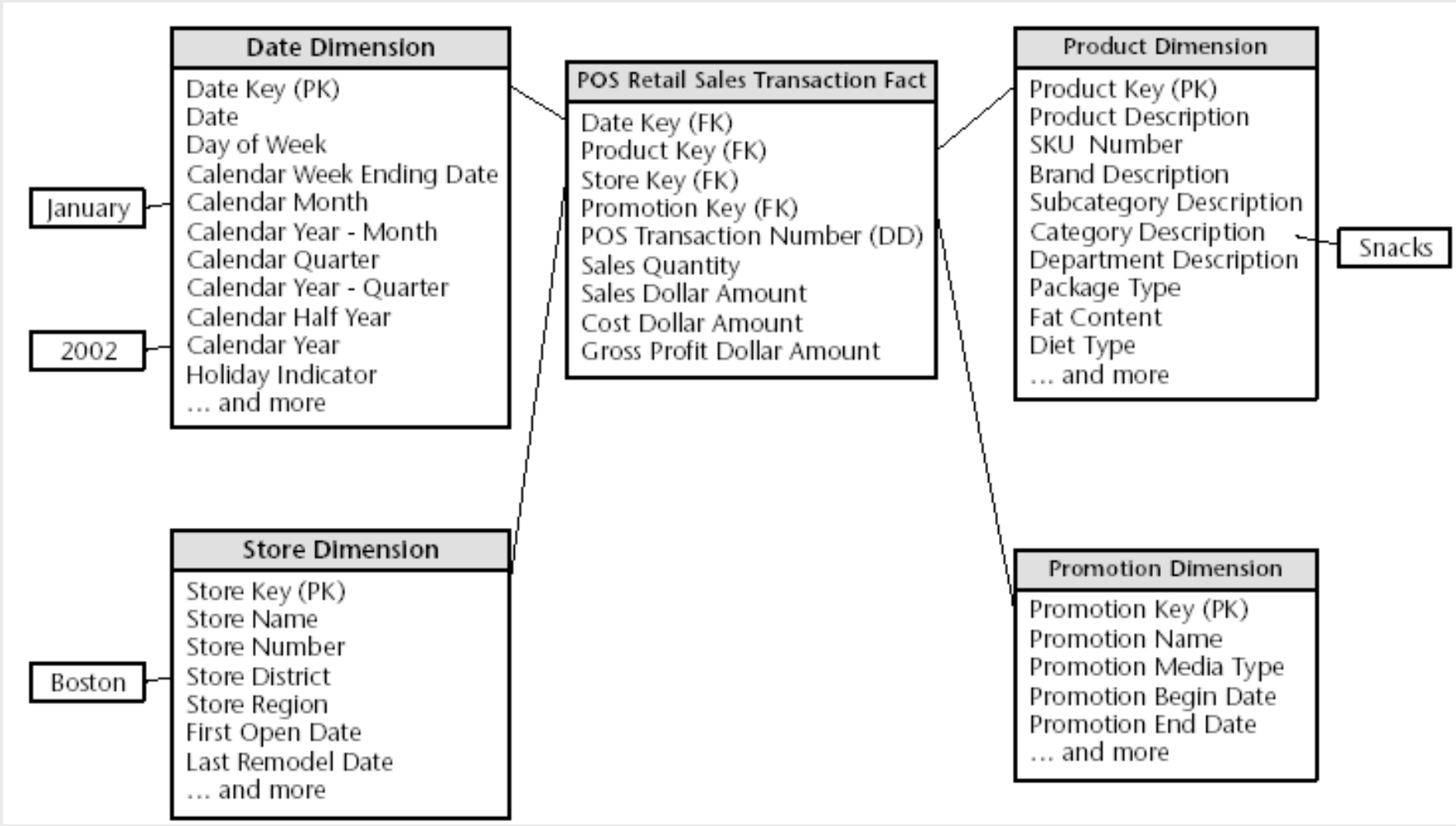


Figure 12.2 Student registration events as a factless fact table.

Surrogate Keys

- **Surrogate Keys**
  - **rather than operational production codes (;natural keys)**
  - **are also called as meaningless keys, integer keys, non-natural keys, artificial keys, synthetic keys**
  - **are integers that are assigned sequentially as needed to populate a dimension**

Retail Case Study: Star-Schema



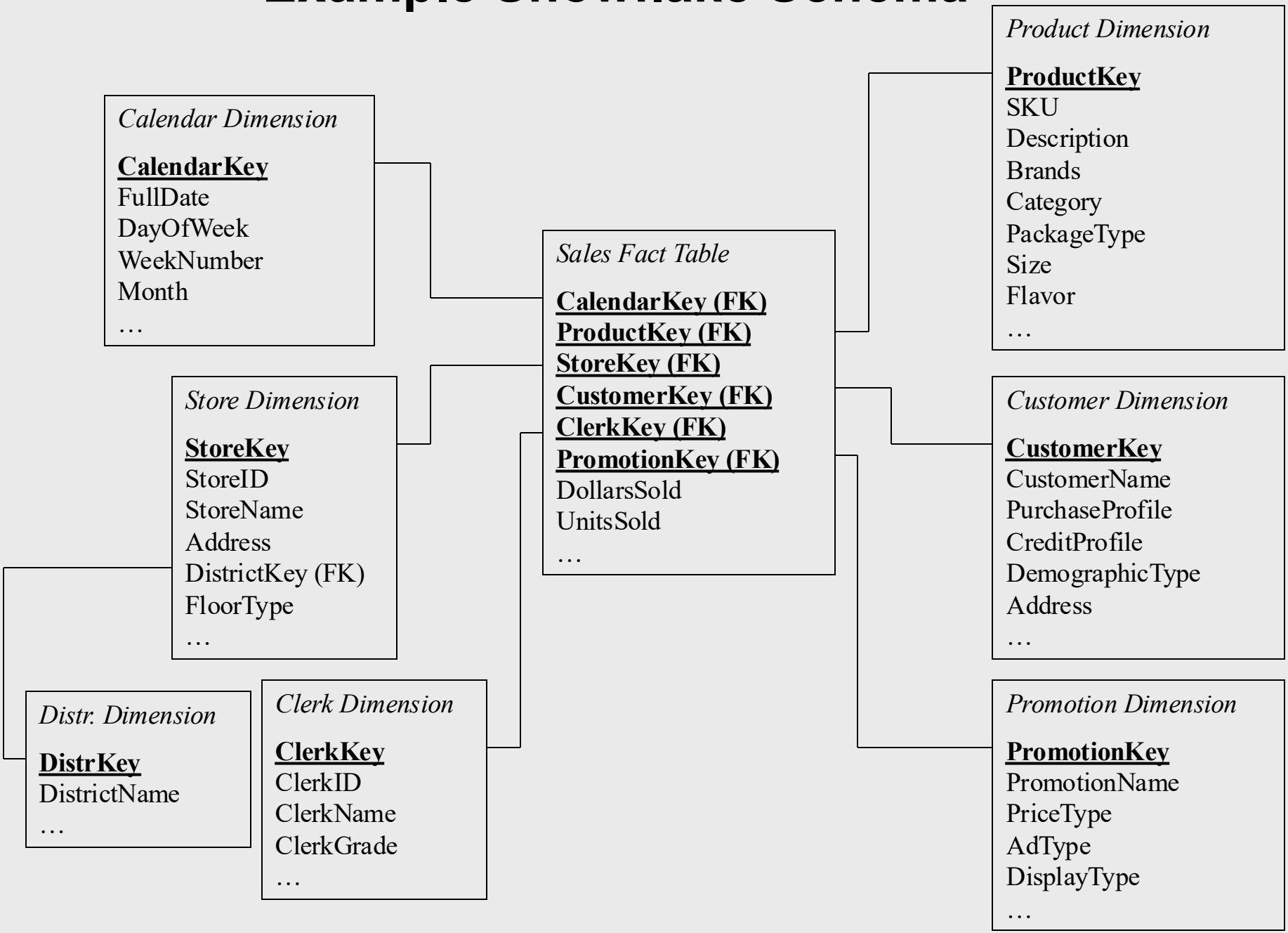
Surrogate Keys *

- **Every join between dimension and fact tables should be based on meaningless integer surrogate keys**
  - **We want to avoid embedding intelligence in the data warehouse keys**
    - because any assumptions that we make eventually may be invalidated
  - **Queries and data access applications should not have any built-in dependency on the keys**
    - because the logic also would be vulnerable to invalidation

Surrogate Keys

- **Surrogate Keys Benefits**
  - **The surrogate keys buffer the data warehouse environment from operational changes**
    - Surrogate keys allow the data warehouse team to integrate data from multiple operational source systems
  - **The surrogate key is as small an integer as possible while ensuring that it will accommodate maximum number of rows in the dimension**
  - **The surrogate keys are used to record dimension conditions that may not have an operational code**
    - "Date to be Determined" or "Date Not Applicable"
  - **Treating the surrogate date key as a date sequence number will allow the fact table to be physically partitioned on the basis of the date key**
    - The partitioning is highly effective because it allows old data and new data to be loaded and indexed without disturbing the rest of the fact table

Example Snowflake Schema



Examples of Fact Tables

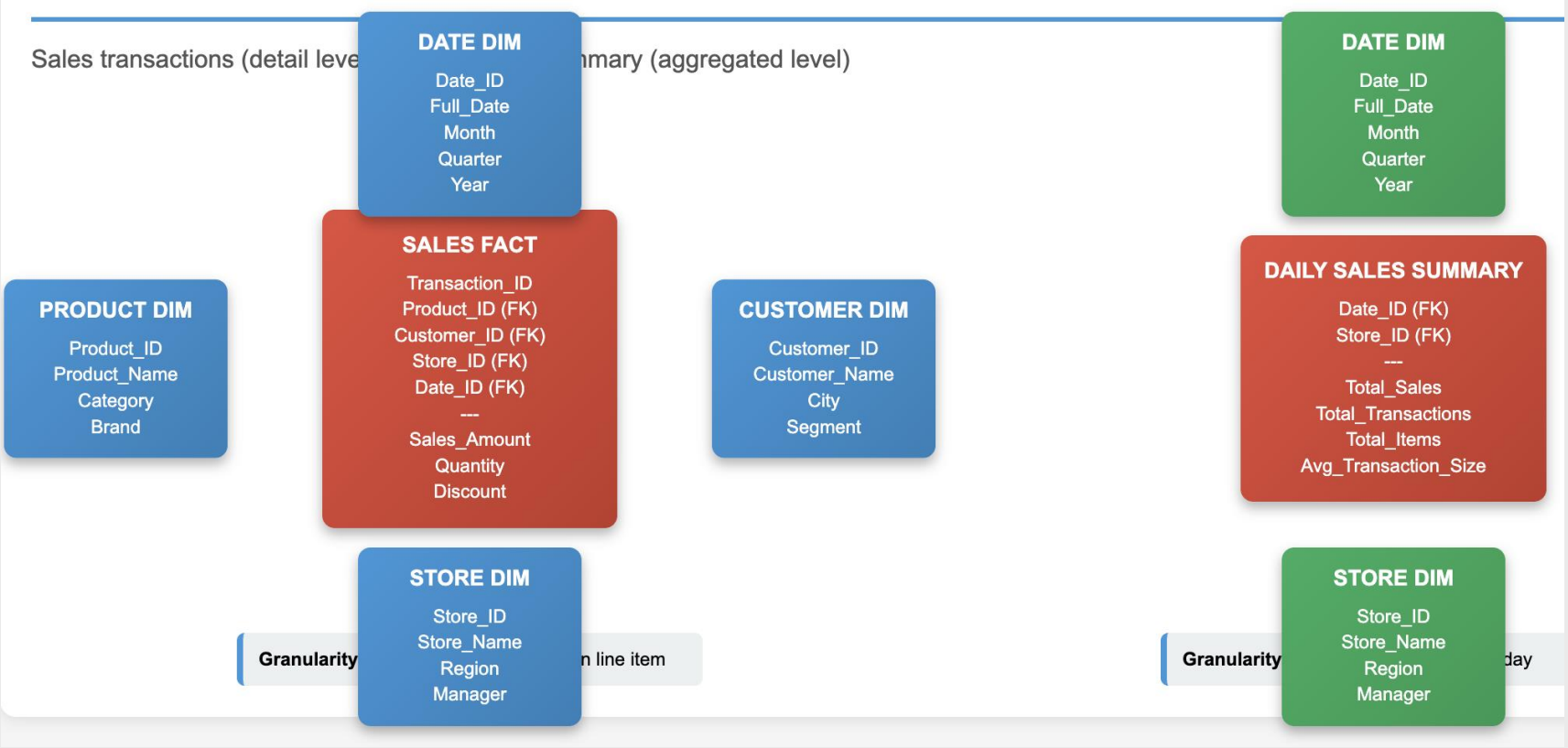
- **Transaction Fact:**
  - Sales transactions with dimensions like Customer, Product, Date, Employee.
- **Periodic Snapshot Fact:**
  - Monthly sales with dimensions like Month, Store, Region.
- **Accumulating Fact:**
  - Order lifecycle with dimensions like
    - Order ID,
    - Customer,
    - Order-Date,
    - Ship-Date,
    - Delivery-Daye.

Multiple fact tables

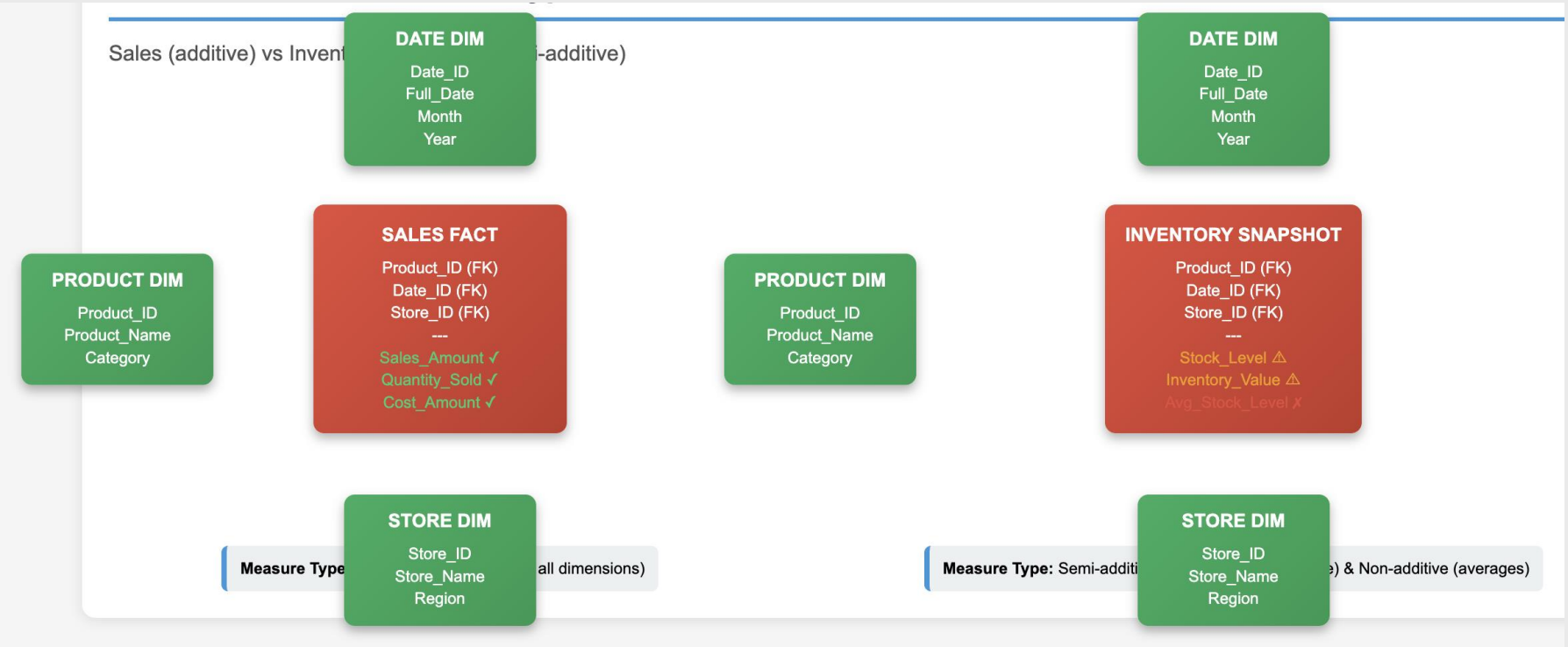
- **Multiple fact tables** is needed because different business processes have fundamentally different characteristics that can't be efficiently combined into a single table.
- **Different Granularities:** different business events occur at **different levels of detail**. For example:
  - **Sales transactions** might be recorded at the individual item level (product, quantity, price per item)
  - **Daily inventory snapshots** are captured once per day per product
  - **Monthly budget** allocations are set at the department level monthly
- **Non-Shared Dimensions:** different business processes often involve completely different dimensional contexts:
  - **Sales facts** use dimensions like Customer, Product, Store, Salesperson
  - **Manufacturing facts** use dimensions like Machine, Operator, Shift, Production Line
  - **HR facts** use dimensions like Employee, Department, Position, Performance Rating
- **Different Measure Types:** various business processes generate fundamentally different types of measurements:
  - **Transactional measures** (sales amount, quantity sold) are additive events
  - **Snapshot measures** (account balance, inventory level) represent states at points in time
  - **Accumulated measures** (order lifecycle from placement to fulfillment) track progress through stages

Different granularities

Granularity must be clearly defined before design

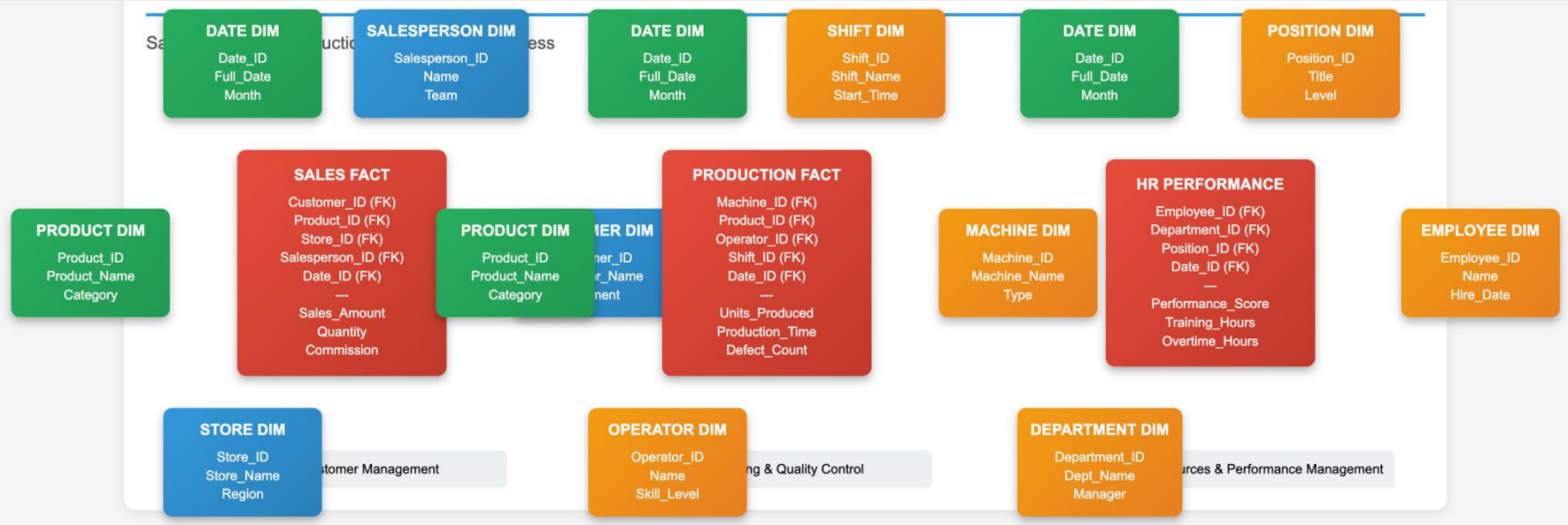


Different types of measures



- Sales_Fact: มี measures แบบ Additive (บวกรวมได้)
  - Sales_Amount = 1,000 บาท
  - Quantity_Sold = 10 ชิ้น
- Inventory_Snapshot_Fact: มี measures แบบ Semi-Additive (บวกรวมได้บางมิติ)
  - Stock_Level = 100 ชิ้น (ไม่สามารถบวกรวมตามเวลาได้)
  - Inventory_Value = 50,000 บาท

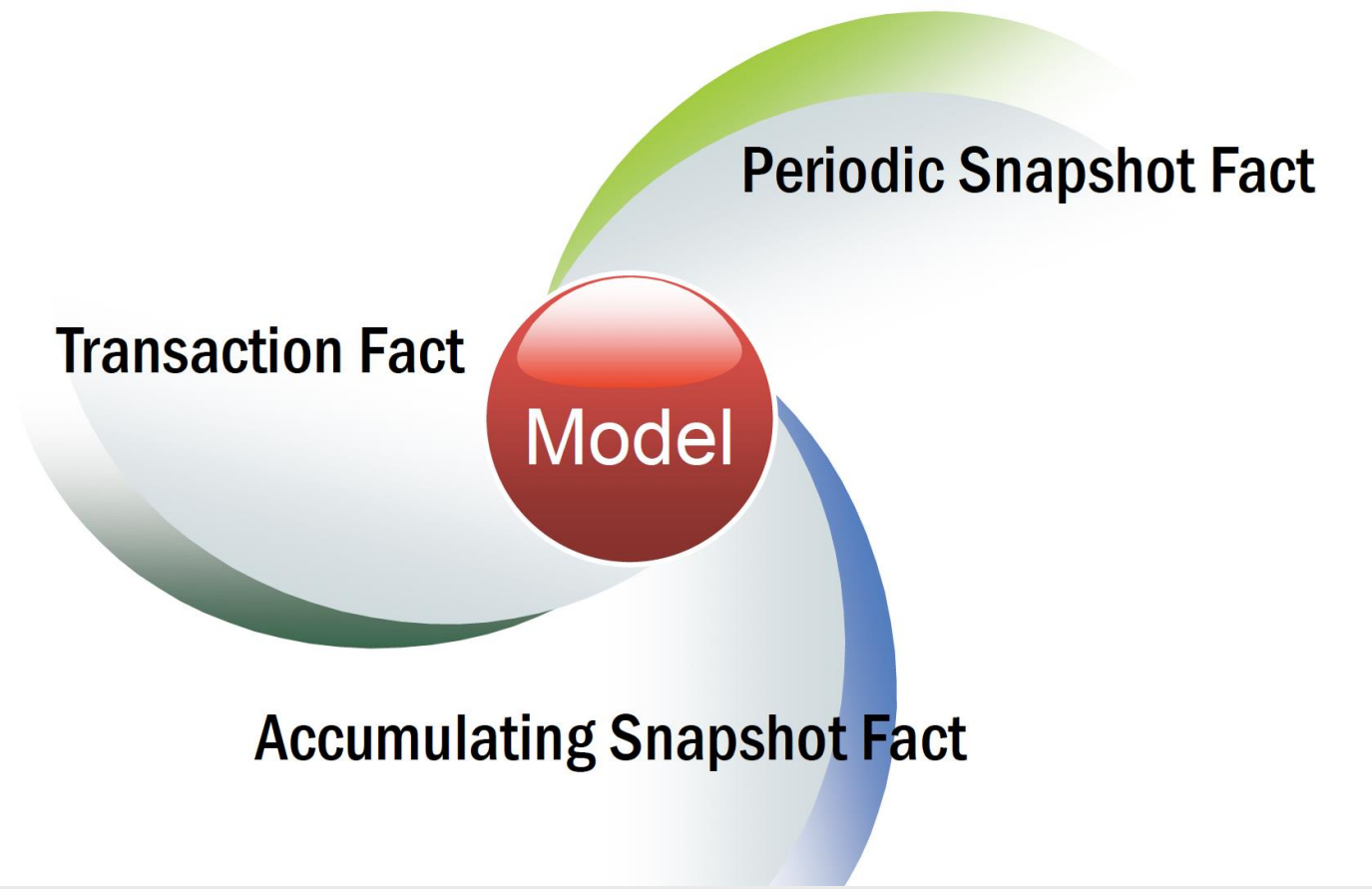
Different business process



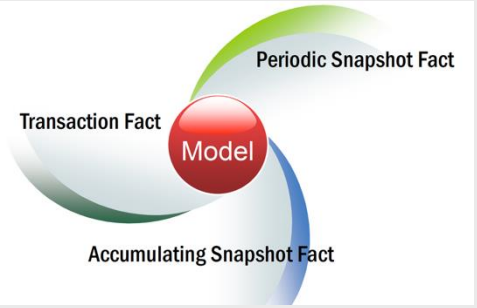
- Blue boxes: Standard dimensions
- Green boxes: Shared dimensions across multiple fact tables
- Orange boxes: Unique dimensions for specific business processes

- **Sales_Fact**: ใช้ Customer, Product, Store, Salesperson
- **Production_Fact**: ใช้ Machine, Operator, Shift, Factory
- **HR_Fact**: ใช้ Employee, Department, Position

Different Types of Fact Tables



Different Types of Fact Tables



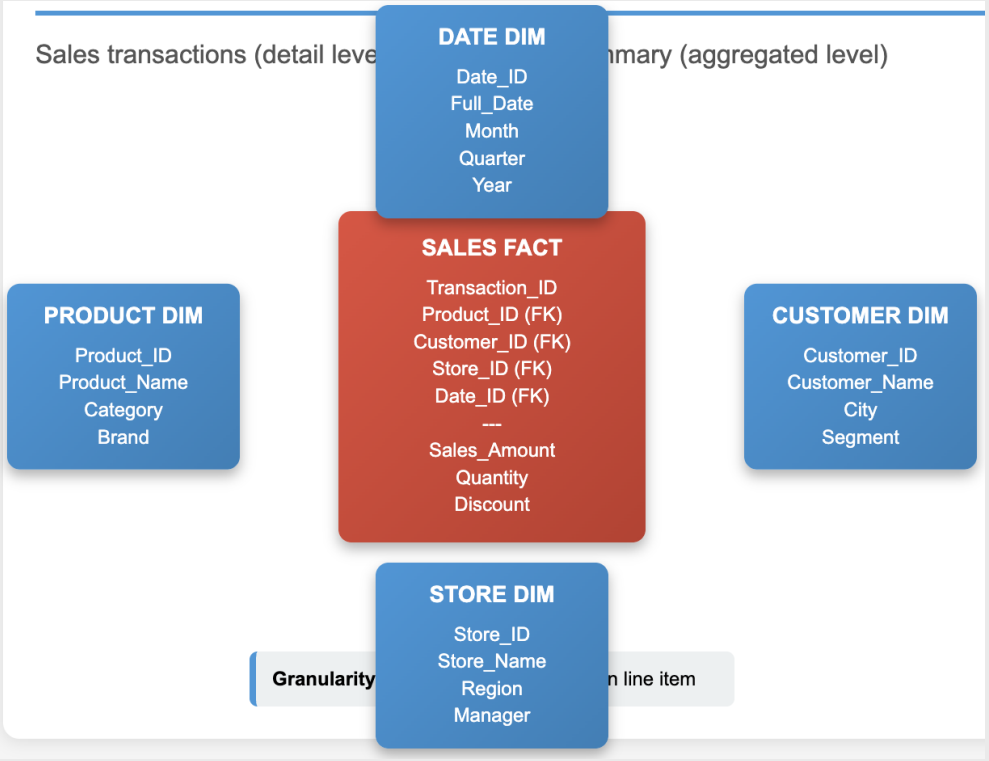
- **Transaction Fact**
  - Captures detailed business events (e.g., sales transaction).
  - บันทึกเหตุการณ์ที่เกิดขึ้นแต่ละครั้ง เช่น การซื้อสินค้า
  - High granularity (ละเอียดที่สุด)
- **Periodic Snapshot Fact**
  - Captures data at regular time intervals (e.g., monthly sales summary).
  - ไม่เก็บรายละเอียดของแต่ละ transaction
  - Time-oriented dimensions (Month, Quarter).
- **Accumulating Snapshot Fact**
  - Tracks lifecycle of a process (e.g., order-to-delivery process).
  - บันทึกการติดตามขั้นตอนของกระบวนการ เช่น order lifecycle
  - มีการอัปเดตค่าเมื่อขั้นตอนต่างๆ สำเร็จ
  - Lot of Date dimensions (OrderDate, ShipDate, etc.).
  - Dimensions for tracking processes such as milestone dates

Examples of Fact Tables

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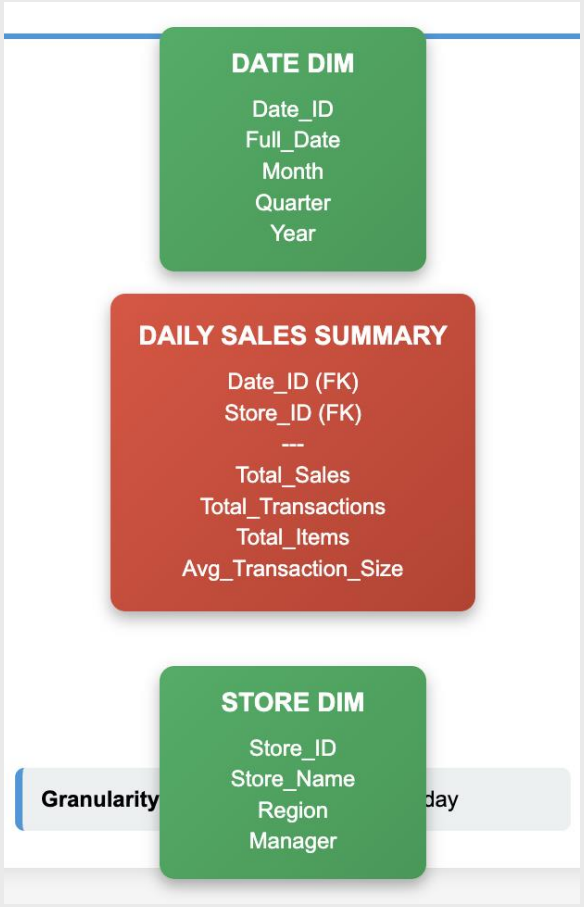
Transaction fact tables

- **Captures**: The most granular events (one row per transaction).
- **Example Business Questions**:
  - What was the total sale from each online order last month?
  - Which products were purchased together most frequently?
  - How much did customer #C123 spend in Q1?
- **Typical Use Case**: Retail sales, e-commerce, financial transactions.



Periodic fact tables

- **Captures**: Periodic summary of data at specific points in time.
- **Example Business Questions**:
  - What was the inventory level for each product at the end of each month?
  - How many open support tickets were there at the start of each day? จำนวนเคสที่ยังไม่ได้ปิด ณ ตอนเช้าของแต่ละวันมีเท่าใด?
  - What was the balance of each bank account every Monday? ยอดเงินในบัญชีของลูกค้าในทุกวันจันทร์เป็นอย่างไร?
- **Typical Use Case**: Inventory tracking, account balance history, HR headcount reporting.



Accumulate Snapshot fact tables

- **Captures**: The lifecycle of a process (single row updated as progress occurs).
- **Example Business Questions**:
  - What is the average time to close a customer support case?
  - เวลาที่ใช้เฉลี่ยในการปิดเคสบริการลูกค้าคือกี่วัน?
  - What percentage of orders were delivered within 3 days?
  - กี่เปอร์เซ็นต์ของคำสั่งซื้อที่จัดส่งได้ภายใน 3 วัน?
  - How long does it typically take to onboard a new employee?
  - ระยะเวลาเฉลี่ยในการอบรมพนักงานใหม่คือเท่าใด?
- **Typical Use Case**: กรณีใช้งาน: ติดตามคำสั่งซื้อ, กระบวนการสมัครงาน, การอนุมัติสินเชื่อ

